

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Eighth Semester B.E (E&C) (Credit System) Degree Examinations**

May - June 2023

19ECE33 / 16ECE34 – BIOMEDICAL INSTRUMENTATION

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1.	a) With a neat block diagram explain basic biomedical instrumentation system.	8	L*2	1	1
	b) With relevant graph, Compare resting and action potential.	6	L2	2	1
	c) Explain the sources of biomedical signals.	6	L2	1	1
2.	a) With a neat diagram, explain the surface electrode and any 4 of it's types briefly.	8	L2	2	1
	b) With a neat block diagram explain Intelligent biomedical instrumentation system.	6	L2	1	1
	c) Explain Micro-electrodes briefly.	6	L2	2	1
3.	a) List and explain the different design specifications of Medical Instruments.	8	L2	2	1
	b) Explain the features and characteristics of ECG and EEG signals.	12	L2	2	1
Unit – II					
4.	a) Explain the role of pressure transducers in medical field? Explain LVDT and strain gauge pressure transducers	10	L2	3	1
	b) Explain Einthoven triangle. Write a note on different ECG lead configurations used in ECG acquisition.	10	L2	4	1
5.	a) Explain transducers and its classification. Explain displacement transducers briefly.	10	L2	3	1
	b) Explain the effects of Artifacts on ECG Recordings and PCG in details.	10	L2	4	1
6.	a) Describe the following transducer used for the measurement of body temperature (1) Thermocouple (2) Thermistor.	10	L2	3	1
	b) With a neat schematic explain instrumentation amplifier and list its applications.	10	L2	4	1
Unit – III					
7.	a) With the help of block diagram, explain the working of MRI Instrumentation system and list the applications.	10	L2	5	1
	b) With the help of neat block diagram of x-ray machine. Explain the characteristic, merit, demerits and applications an x-ray machine.	10	L2	5	1
8.	a) Explain computer tomography system and its major subsystems.	10	L2	5	1
	b) With the help of neat block diagram of Echocardiograph, explain Echocardiography and types of echocardiograms.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
